

Sec. 6.1. Cryptography

Task 1- Introduction

Cryptography lays the foundation for our digital world. While networking protocols have made it possible for devices spread across the globe to communicate, cryptography has made it possible to trust this communication.

Task 2 - Importance of Cryptography

Cryptography's ultimate purpose is to ensure *secure communication in the presence of adversaries*. The term secure includes confidentiality and integrity of the communicated data. Cryptography can be defined as the practice and study of techniques for secure communication and data protection where we expect the presence of adversaries and third parties. In other words, these adversaries should not be able to disclose or alter the contents of the messages.

Answer the questions below

| What is the standard required for handling credit card information?

PCI DSS

Task 3 - Plaintext to Ciphertext

- **Plaintext** is the original, readable message or data before it's encrypted. It can be a document, an image, a multimedia file, or any other binary data.
- **Ciphertext** is the scrambled, unreadable version of the message after encryption. Ideally, we cannot get any information about the original plaintext except its approximate size.
- **Cipher** is an algorithm or method to convert plaintext into ciphertext and back again. A cipher is usually developed by a mathematician.
- **Key** is a string of bits the cipher uses to encrypt or decrypt data. In general, the used cipher is public knowledge; however, the key must remain secret

unless it is the public key in asymmetric encryption. We will visit asymmetric encryption in a later task.

- **Encryption** is the process of converting plaintext into ciphertext using a cipher and a key. Unlike the key, the choice of the cipher is disclosed.
- **Decryption** is the reverse process of encryption, converting ciphertext back into plaintext using a cipher and a key. Although the cipher would be public knowledge, recovering the plaintext without knowledge of the key should be impossible (infeasible).

Answer the questions below

| What do you call the encrypted plaintext? ciphertext

| What do you call the process that returns the plaintext? decryption

Task 4 - Types of Encryption

The two main categories of encryption are **symmetric** and **asymmetric**.

Symmetric encryption, also known as **symmetric cryptography**, uses the same key to encrypt and decrypt the data, as shown in the figure below. Keeping the key secret is a must; it is also called **private key cryptography**. Furthermore, communicating the key to the intended parties can be challenging as it requires a secure communication channel. Maintaining the secrecy of the key can be a significant challenge, especially if there are many recipients.

There are many more symmetric encryption ciphers used in various applications. Examples of symmetric encryption are DES (Data Encryption Standard), 3DES (Triple DES) and AES (Advanced Encryption Standard).

- **DES** was adopted as a standard in 1977 and uses a 56-bit key. With the advancement in computing power, in 1999, a DES key was successfully broken in less than 24 hours, motivating the shift to 3DES.
- **3DES** is DES applied three times; consequently, the key size is 168 bits, though the effective security is 112 bits. 3DES was more of an ad-hoc solution when DES was no longer considered secure. 3DES was deprecated in 2019 and should be replaced by AES; however, it may still be found in some legacy systems.

- **AES** was adopted as a standard in 2001. Its key size can be 128, 192, or 256 bits.

Asymmetric Encryption: Unlike symmetric encryption, which uses the same key for encryption and decryption, **asymmetric encryption** uses a pair of keys, one to encrypt and the other to decrypt, as shown in the illustration below. To protect confidentiality, asymmetric encryption or **asymmetric cryptography** encrypts the data using the public key; hence, it is also called **public key cryptography**.

Examples are RSA, Diffie-Hellman, and Elliptic Curve cryptography (ECC). The two keys involved in the process are referred to as a **public key** and a **private key**. Data encrypted with the public key can be decrypted with the private key. Your private key needs to be kept private, hence the name.

Asymmetric encryption tends to be slower, and many asymmetric encryption ciphers use larger keys than symmetric encryption. For example, RSA uses 2048-bit, 3072-bit, and 4096-bit keys; 2048-bit is the recommended minimum key size. Diffie-Hellman also has a recommended minimum key size of 2048 bits but uses 3072-bit and 4096-bit keys for enhanced security. On the other hand, ECC can achieve equivalent security with shorter keys. For example, with a 256-bit key, ECC provides a level of security comparable to a 3072-bit RSA key.

Asymmetric encryption is based on a particular group of mathematical problems that are easy to compute in one direction but extremely difficult to reverse. In this context, extremely difficult means practically infeasible. For example, we can rely on a mathematical problem that would take a very long time, for example, millions of years, to solve using today's technology.

Symmetric encryption is a method in which the same key is used for both encryption and decryption. Consequently, this key must remain secure and never be disclosed to anyone except the intended party. **Asymmetric encryption** is a method that uses two different keys: a public key for encryption and a private key for decryption.

Answer the questions below

| Should you trust DES? Nay

| When was AES adopted as an encryption standard? 2001

